LABORATORY · Lab · 26 hours

Experiment 5: Root Locus, Breakaway and Imaginary-Axis Crossover



Control Systems Laboratory (22EEG53)

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Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Plot the root locus of a given transfer function
- Verify the breakaway point analytically and in MATLAB
- Find the imaginary-axis crossing and the marginal gain
- Relate the gain to damping

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

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Aim and requirements

Aim: to draw the root locus for a given transfer function and verify the breakaway point and imaginary-axis crossover point using MATLAB.

Software: MATLAB with the Control System Toolbox (Simulink where required).

Hardware: Personal computer.

System: $G(s)H(s) = K/[s(s + 1)(s + 2)]$.

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Hand calculations

Asymptotes 60°, 180°, 300°

centroid σA = −1

Breakaway: $3s^2 + 6s + 2 = 0$

s = −0.423 (on locus)

Routh: K = 6

jω crossing at ±j1.414

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Procedure

1

Define G

G = tf(1, [1 3 2 0]).

2

Plot

rlocus(G); sgrid or zeta lines.

3

Breakaway

Click on the locus or compute roots of d/ds of the characteristic polynomial.

4

Crossover

rlocfind or margin: K at the jω-axis intersection.

Check with Routh: marginal gain K = 6.

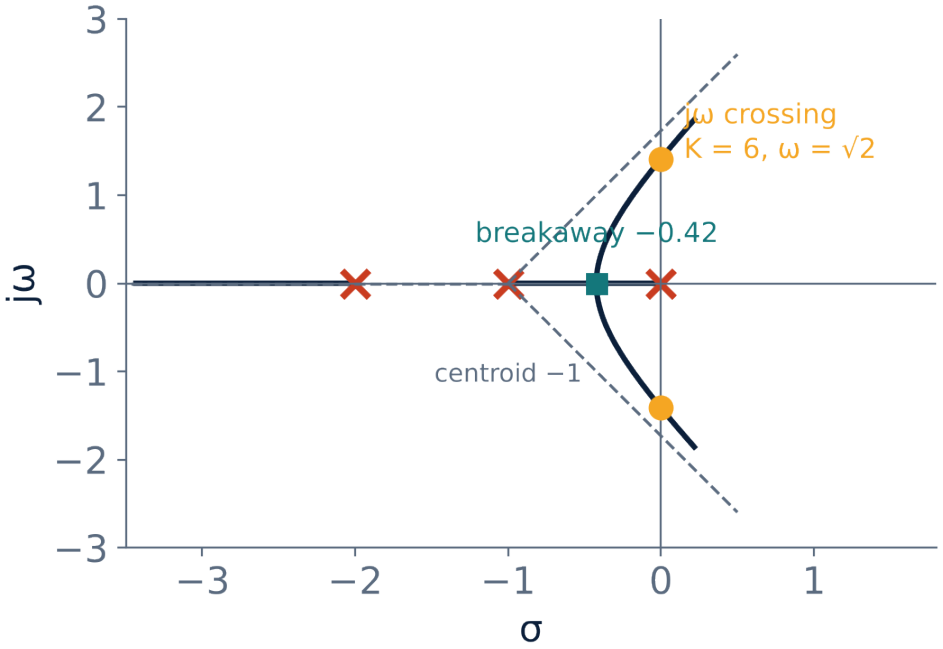
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MATLAB program

```
G = tf(1, [1 3 2 0]);
rlocus(G); sgrid; grid on
% breakaway: roots of dK/ds = 0, K = -s(s+1)(s+2)
p  = conv([1 0], conv([1 1], [1 2]));
ds = roots(polyder(p))                % -0.4226, -1.5774
K_b = -polyval(p, ds(1))              % gain at breakaway = 0.385
% imaginary axis crossing
[Gm, ~, wpc] = margin(G)              % Gm = 6, wpc = 1.414
K_marginal = Gm
```

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Expected root locus



- Breakaway at −0.423 (K = 0.385).
- Crossing at ±j1.414 for K = 6.
- Asymptotes at ±60°, 180°.

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Observation table

Quantity	Hand calculation	MATLAB
Breakaway point	−0.423	
Gain at breakaway	0.385	
Imaginary-axis crossing	±j1.414	
Marginal gain K	6	

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Key takeaways

- rlocus plots the locus; sgrid shows ζ and ωn lines
- Breakaway: dK/ds = 0 → −0.423
- Marginal stability at K = 6, ω = 1.414 rad/s
- MATLAB results agree with Routh and the rules

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Practice questions

Q1. List the steps to sketch the locus of K/[s(s+1)(s+2)].

Q2. Write the MATLAB commands for the locus.

Q3. Find K for ζ = 0.5 using rlocfind.

Q4. What happens for K > 6?